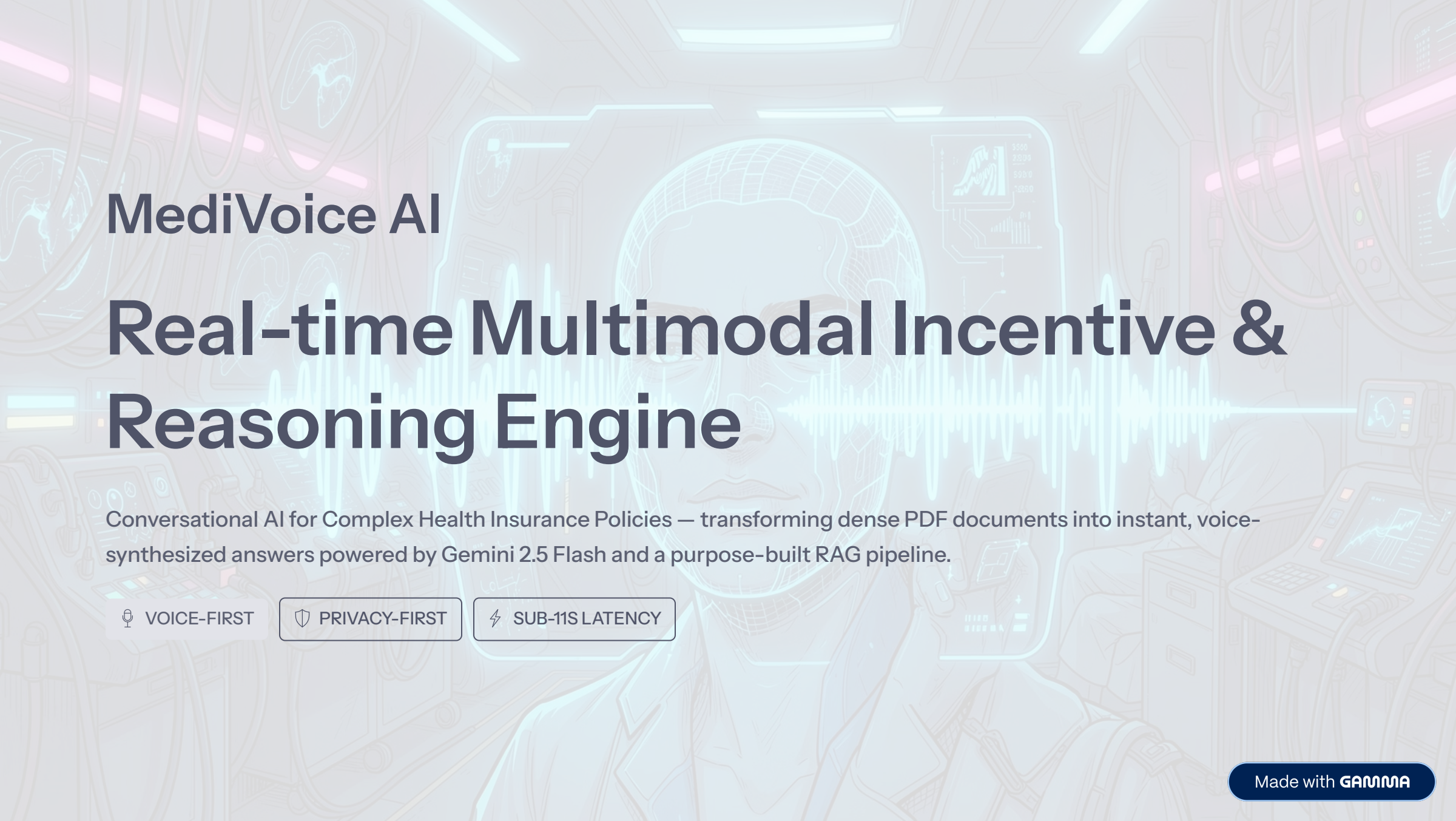




MediVoice AI

Real-time Multimodal Incentive & Reasoning Engine

Conversational AI for Complex Health Insurance Policies — transforming dense PDF documents into instant, voice-synthesized answers powered by Gemini 2.5 Flash and a purpose-built RAG pipeline.

 VOICE-FIRST

 PRIVACY-FIRST

 SUB-11S LATENCY

Made with **GAMMA**



The Problem: Policy Complexity at Scale

Health insurance policies from providers like HDFC Ergo or Star Health routinely exceed **80 pages** of dense, legally-worded PDFs. For the average policyholder, navigating co-pays, exclusions, sub-limits, and pre-authorization rules is nearly impossible without expert assistance.

Document Density

80+ page policies written in legal terminology that no layperson can parse quickly or accurately.

Call Center Overload

High volumes of inbound calls for basic policy queries drive up operational costs and frustrate customers.

Slow Resolution

Customers wait days for email responses or sit on hold for hours — a broken experience in 2025.

Poor Outcomes

Misunderstood exclusions and co-pay clauses lead to claim rejections, eroding trust in insurers.

The Solution: MediVoice AI

MediVoice AI is an **end-to-end multimodal voice agent** that lets users upload their specific policy PDF and simply **talk to it** using a microphone. The system instantly retrieves accurate, cited answers and delivers them as natural, voice-synthesized speech.

No chat interfaces. No scrolling through documents. Just ask a question in plain English and receive a precise, policy-grounded response in seconds.

→ Upload Your Policy

Drag-and-drop any health insurance PDF — the system ingests it in real time.

→ Ask Anything by Voice

Tap the microphone and speak naturally: "What's my co-pay for outpatient visits?"

→ Get Instant, Cited Answers

Receive a voice response backed by document citations and a transparent reasoning chain.



Core Architecture

MediVoice AI is built on a lean, modern stack designed for CPU-only cloud deployment. Every component was selected to minimize latency and cost while maximizing accuracy and scalability on Hugging Face Spaces.



FastAPI (Python)

Async REST backend handling all HTTP and WebSocket connections with sub-millisecond routing overhead.



Gemini 2.5 Flash

Google's multimodal LLM serves as the reasoning engine — processing audio, text, and vector context simultaneously.



ChromaDB

In-memory vector database for semantic similarity search over policy document chunks. Zero persistent storage footprint.



gTTS

Google Text-to-Speech synthesizes natural voice responses from the LLM's conversational script output.



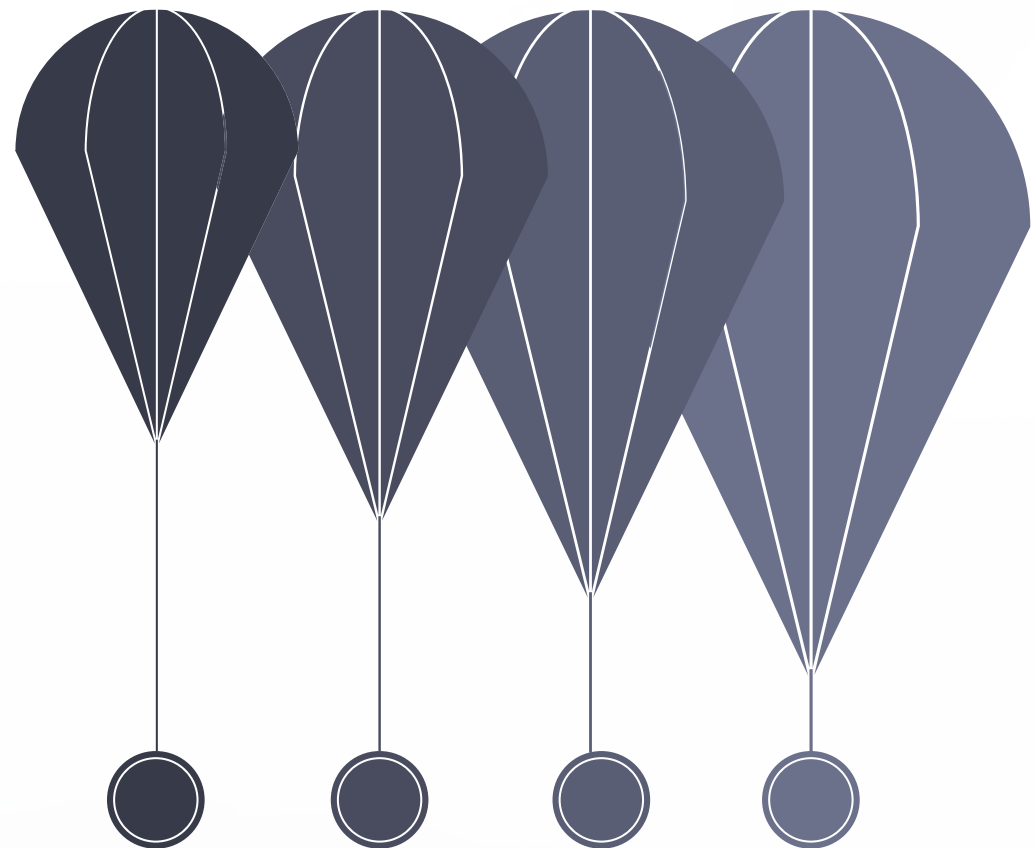
Hugging Face Spaces

Dockerized container deployment on CPU-only infrastructure — eliminating GPU costs entirely while maintaining sub-11s response times.

The RAG Pipeline: Dynamic Knowledge Ingestion

When a user uploads a policy PDF through the UI, MediVoice AI immediately begins a **zero-latency ingestion pipeline**. The document is chunked into semantically meaningful segments, each vectorized using Gemini's embeddings, and stored in an ephemeral ChromaDB instance.

-  **Upload PDF**
User submits policy document via UI
-  **Chunk & Vectorize**
Split into semantic segments and embed
-  **Store in ChromaDB**
Load vectors into ephemeral in-memory DB
-  **Query & Retrieve**
Semantic search returns relevant chunks



This ephemeral architecture is a deliberate design choice: **cloud storage wipes clean on every server restart**, ensuring no policy data persists between sessions. This approach guarantees user privacy by design and eliminates long-term storage costs — a critical consideration for a SaaS product handling sensitive health insurance documents.

 **Privacy by Design:** No policy document is ever stored persistently. All vector data is purged automatically on container restart, ensuring GDPR-compliant, session-scoped data handling.

Real-Time Voice Processing: The Engineering Pivot

The Original Approach (Abandoned)

Initial prototypes used **Whisper** (OpenAI's local STT model) for speech-to-text transcription. This required:

- Heavy GPU resources for real-time inference
- Audio file upload to Google's file queue
- Significant latency overhead (20–40s per turn)
- Complex async file management on CPU servers

The result: an architecture too slow and expensive for production.

The Gemini 2.5 Flash Solution

MediVoice AI pivoted to streaming **raw audio "inline bytes"** directly to Gemini 2.5 Flash's native multimodal audio endpoint. This bypasses the Google File upload queue entirely.

- **Zero local STT model** — transcription handled natively by Gemini
- **CPU-only compatible** — no GPU required on Hugging Face Spaces
- **Near-instant transcription** — audio processed inline during the same API call
- **End-to-end latency under 11 seconds** — from microphone tap to voice output

Hallucination Guardrails: Structured Output Engineering

Preventing LLM hallucination in a regulated domain like health insurance is non-negotiable. MediVoice AI enforces a **strict dual-output JSON schema** that separates the AI's internal reasoning from its conversational output.



Strict JSON Schema Enforcement

Gemini is prompted to return **only valid JSON** — no free-form text. The schema has two top-level keys: `reasoning_chain` (internal citations) and `voice_script` (user-facing response). Any deviation triggers a retry.



Reasoning Chain (Internal)

Contains exact document chunk citations, vector similarity scores, and the logical steps the model took to arrive at its answer. This is visible in the UI's audit panel but **never spoken aloud**.



Voice Script (External)

A natural, conversational response derived from the reasoning chain — rewritten to sound human, not robotic. The prompt explicitly instructs Gemini to **avoid jargon** and use plain English.



Context-Bound Responses

System prompts instruct Gemini to **refuse answers** when the vector context does not contain sufficient information. The model cannot "guess" — it must cite its source chunks or decline.

Telemetry & Unit Economics

MediVoice AI's UI includes a **live telemetry dashboard** that exposes the exact operational cost and performance of every interaction. This is a deliberate PM-focused feature — engineering leaders and recruiters can see the system's efficiency in real time.

11s

Max End-to-End Latency

From microphone tap to voice output — measured across all query types on CPU-only infrastructure.

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API Cost Per Turn

Estimated cost per Gemini 2.5 Flash API call including audio transcription and text generation.

98%

RAG Confidence Threshold

Minimum vector similarity score required before the system will generate an answer — reducing hallucination risk.

0

Persistent Storage Cost

Ephemeral ChromaDB architecture means zero long-term database costs — all data purges on restart.

Token Tracking

The dashboard displays exact **input and output token counts** per turn, enabling precise cost attribution and LLM optimization. Engineering teams can identify expensive query patterns and optimize chunk sizes accordingly.

RAG Confidence Scores

Each response includes a **vector similarity score** shown in the audit panel. Low-confidence answers are flagged, giving users and developers visibility into when the system is operating near the edge of its knowledge base.

The User Journey: Demo Flow

MediVoice AI's interaction model is intentionally simple — three steps from upload to answer. The demo flow is designed to showcase the full capability of the system in under 60 seconds.



Step 1: Sync Data

User uploads their health insurance policy PDF via the drag-and-drop UI. The system immediately chunks, vectorizes, and stores the document in the ephemeral ChromaDB instance. A confirmation badge indicates the knowledge base is ready.



Step 2: Query

User taps the microphone button and asks a natural-language question: *"What is my co-pay for hospitalization?"* Raw audio bytes stream directly to Gemini 2.5 Flash — no intermediate STT layer, no file upload queue.



Step 3: Audit

The UI displays two panels simultaneously: the **internal reasoning chain** (citations, confidence scores, logical steps) and the **voice response** (natural speech synthesized via gTTS). Users can review exactly how the answer was derived before accepting it.

Conclusion & Call to Action

MediVoice AI demonstrates that **complex, regulated domains** like health insurance can be made accessible through thoughtful AI architecture. By combining Gemini 2.5 Flash's multimodal capabilities with a privacy-first RAG pipeline, strict hallucination guardrails, and CPU-only deployment, the system delivers enterprise-grade accuracy at consumer-grade cost.

Speed

Sub-11-second end-to-end voice responses on CPU-only cloud infrastructure.

Accuracy

Strict JSON output schemas and RAG confidence thresholds eliminate hallucination risk.

Scalable

Ephemeral, Dockerized architecture on Hugging Face Spaces — zero persistent state, zero GPU costs.

Try the Live Demo

Experience MediVoice AI in action. Upload a sample policy PDF and ask your first question in under 60 seconds.

[Launch Live Demo](#)

View the Source Code

The full codebase is open source on GitHub — including the FastAPI backend, RAG pipeline, prompt engineering templates, and telemetry dashboard.

[View on GitHub](#)

- 📄 **Built for engineering leaders and tech recruiters:** MediVoice AI is a production-ready demonstration of modern LLM engineering — RAG pipelines, multimodal APIs, structured output enforcement, and real-time telemetry, all in a single deployable container.